

Lecture 41 – Questions and Answers (Ensemble Methods, Bagging, Boosting, Random Forests)

I. Ensemble Learning Fundamentals (1–10)

1. **Q:** What is the core concept behind ensemble learning?
A: Ensemble learning combines multiple models (weak learners) to improve predictive performance and generalization.
 2. **Q:** Why is ensemble learning generally more powerful than a single model?
A: Because it reduces variance, bias, and the likelihood of overfitting by averaging or voting among several models.
 3. **Q:** What is a *weak learner* in ensemble methods?
A: A model that performs slightly better than random guessing.
 4. **Q:** What is the difference between *homogeneous* and *heterogeneous* ensembles?
A: Homogeneous ensembles use the same type of model (e.g., multiple decision trees), while heterogeneous ones use different models.
 5. **Q:** What are the three main categories of ensemble methods?
A: Bagging, Boosting, and Stacking.
 6. **Q:** What does Bagging stand for?
A: Bootstrap Aggregating.
 7. **Q:** What is the purpose of the bootstrap step in Bagging?
A: It generates different training datasets by sampling with replacement from the original data.
 8. **Q:** What is the purpose of the aggregation step in Bagging?
A: To combine multiple predictions (by averaging or voting) into one final output.
 9. **Q:** What is the main advantage of Bagging?
A: It reduces variance without increasing bias.
 10. **Q:** Which base learner is most commonly used in Bagging?
A: Decision Trees.
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II. Bagging and Random Forests (11–20)

11. **Q:** How does Random Forest improve upon Bagging?
A: By introducing additional randomness — selecting random subsets of features for each tree.
 12. **Q:** What does the “forest” in Random Forest represent?
A: An ensemble of decision trees trained on random subsets of data and features.
 13. **Q:** What metric is typically used to split nodes in Random Forest regression?
A: Mean Squared Error (MSE).
 14. **Q:** What metric is typically used to split nodes in Random Forest classification?
A: Gini Impurity or Entropy.
 15. **Q:** What are *Out-of-Bag (OOB)* samples?
A: Data points not selected in the bootstrap sample, used for internal validation.
 16. **Q:** How is the OOB error estimated?
A: By averaging prediction errors for each data point using only the trees that did not see it during training.
 17. **Q:** What percentage of data is typically OOB in each bootstrap sample?
A: Approximately 36.8%.
 18. **Q:** Why is Random Forest resistant to overfitting?
A: Because of feature randomness and averaging across many uncorrelated trees.
 19. **Q:** What parameter controls the number of trees in a Random Forest?
A: The hyperparameter `n_estimators`.
 20. **Q:** What is one disadvantage of Random Forest compared to a single tree?
A: It is less interpretable.
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III. Boosting Methods (21–35)

21. **Q:** What is the main idea behind Boosting?
A: Sequentially combining weak learners so that each new learner focuses on correcting previous errors.

22. **Q:** What is AdaBoost short for?
A: Adaptive Boosting.
23. **Q:** What type of base learner is commonly used in AdaBoost?
A: Decision stumps (shallow trees with one split).
24. **Q:** How does AdaBoost assign weights to data points?
A: It increases the weights of misclassified points to make them more influential in the next iteration.
25. **Q:** How are final predictions made in AdaBoost?
A: By taking a weighted majority vote of all weak learners.
26. **Q:** What is the main difference between Bagging and Boosting?
A: Bagging trains models independently; Boosting trains them sequentially.
27. **Q:** What is Gradient Boosting?
A: A boosting method where new models are trained to predict the residual errors of prior models.
28. **Q:** What is the objective function minimized in Gradient Boosting?
A: A differentiable loss function, such as MSE for regression or Log Loss for classification.
29. **Q:** How does XGBoost differ from traditional Gradient Boosting?
A: XGBoost includes regularization and parallel computation for improved performance and speed.
30. **Q:** What does “shrinkage” or “learning rate” control in Boosting algorithms?
A: The contribution of each new weak learner to prevent overfitting.
31. **Q:** What is LightGBM?
A: A gradient boosting framework using tree-based learning with histogram-based splitting.
32. **Q:** What is CatBoost?
A: A boosting algorithm that efficiently handles categorical features natively.
33. **Q:** What is one key disadvantage of Boosting methods?
A: They can overfit if too many weak learners are added.
34. **Q:** What is the role of *early stopping* in Boosting?
A: To prevent overfitting by halting training when validation performance stops

improving.

35. **Q:** What is a typical hyperparameter that controls the number of boosting stages?

A: `n_estimators` or `num_boost_round`.

IV. Evaluation and Practical Use (36–50)

36. **Q:** What performance metric is typically used for regression ensemble models?

A: Mean Squared Error (MSE) or Root Mean Squared Error (RMSE).

37. **Q:** What metric is typically used for classification ensemble models?

A: Accuracy, F1-score, or AUC-ROC.

38. **Q:** What is model diversity, and why is it important in ensembles?

A: Diversity means models make different errors — it improves ensemble robustness.

39. **Q:** How can model diversity be increased?

A: By using different algorithms, random seeds, or subsets of data/features.

40. **Q:** Why are ensemble methods often used in Kaggle competitions?

A: They usually outperform single models by combining complementary strengths.

41. **Q:** What is stacking in ensemble learning?

A: Combining predictions from multiple base models using a meta-model.

42. **Q:** How does blending differ from stacking?

A: Blending uses a holdout set for combining models, while stacking uses cross-validation.

43. **Q:** Why is interpretability often reduced in ensemble models?

A: Because the combined decision process of many models is complex and non-linear.

44. **Q:** How can interpretability be improved in ensembles?

A: By using SHAP or feature importance analysis.

45. **Q:** What is the trade-off between bias and variance in ensemble learning?

A: Ensembles reduce variance but may slightly increase bias.

46. **Q:** Why is feature scaling often unnecessary for tree-based ensembles?

A: Because trees use threshold-based splits that are scale-invariant.

47. **Q:** What happens if the base learners in Bagging are highly correlated?
A: The ensemble's performance gain will be small.
48. **Q:** What is the role of random seed initialization in ensemble training?
A: To ensure reproducibility of results.
49. **Q:** What is the computational downside of large ensembles?
A: They require more training time and memory.
50. **Q:** Why are ensembles considered "robust" models?
A: Because averaging or voting across diverse models stabilizes predictions and reduces outlier effects.